Member 1: Literature, research design, and scratch baseline

This person is responsible for understanding the academic foundation of the project.

They will:

- ☐ Find papers about transfer learning and feature transferability.
- ☐ Explain why fine-tuning depth may affect performance.
- ☐ Help finalize the research question and hypotheses.
- ☐ Implement ResNet-50 with randomly initialized weights.
- ☐ Train the scratch model using the agreed data-size settings.
- ☐ Compare scratch performance against pretrained models.
- ☐ Write the problem statement, motivation, and related-work sections.
- ☐ Present the motivation and baseline method.

Their main contribution is establishing what the project is studying and providing the non-transfer-learning baseline.

Member 2: Dataset and data-size experiments

This person owns the data pipeline and the main extension to the project.

They will:

- ☐ Load the Flowers-102 dataset.
- ☐ Verify the official training, validation, and test splits.
- ☐ Implement image resizing, normalization, and augmentation.
- ☐ Create reproducible subsets using a fixed random seed.
- ☐ Make sure reduced datasets are class-balanced.
- ☐ Prepare the 25%, 50%, and 100% training-data conditions.
- ☐ Visualize sample images and confirm that labels are correct.
- ☐ Help run the data-size experiments.
- ☐ Write the dataset and preprocessing sections.
- ☐ Present how the data-size comparison was designed.

Their main contribution is testing whether transfer learning becomes more useful when less labeled data is available.

Member 3: Frozen and selective fine-tuning

This person owns two of the transfer-learning strategies.

They will implement:

- **Frozen feature extraction**
  - Freeze the pretrained ResNet layers.
  - Train only the new Flowers-102 classifier.
- **Selective fine-tuning**
  - Freeze the early and middle layers.
  - Fine-tune the final ResNet stage, `layer4`, and the classifier.

They will also:

- ☐ Count trainable parameters.
- ☐ Compare how many layers are updated.
- ☐ Track accuracy, F1-score, and training time.
- ☐ Save the best validation model.
- ☐ Analyze whether adapting higher-level features improves results.
- ☐ Write the methodology section explaining the fine-tuning strategies.
- ☐ Present the difference between frozen and selective fine-tuning.

Their main contribution is studying how much of the pretrained network needs to change.

Member 4: Full fine-tuning and optimization

This person owns the most flexible and computationally expensive method.

They will:

- ☐ Implement full fine-tuning of ResNet-50.
- ☐ Choose and justify the learning rate.
- ☐ Compare learning-rate choices if time permits.
- ☐ Monitor training time and computational requirements.
- ☐ Track training versus validation performance.
- ☐ Investigate whether full fine-tuning causes overfitting.
- ☐ Save the best-performing model.
- ☐ Compare full fine-tuning against frozen and selective fine-tuning.
- ☐ Write the training and optimization details.
- ☐ Present the strengths and weaknesses of updating the entire network.

Their main contribution is studying the tradeoff between model flexibility, performance, training cost, and overfitting.

Member 5: Evaluation, comparison, and reproducibility

This person creates the shared system used to compare every model fairly.

They will implement:

- [ ] Accuracy
- [ ] Macro-averaged precision
- [ ] Macro-averaged recall
- [ ] Macro-averaged F1-score
- [ ] Confusion matrices
- [ ] Training and validation curves
- [ ] Correct and incorrect prediction visualizations
- [ ] Training-time comparisons
- [ ] Trainable-parameter comparisons
- [ ] Final summary tables

They will also:

- [ ] Make sure every model uses the same test set.
- [ ] Verify that no test results are used for model selection.
- [ ] Run selected experiments with another random seed if possible.
- [ ] Organize the final results.
- [ ] Identify which classes are easiest and hardest to classify.
- [ ] Write the evaluation, results, and conclusion sections.
- [ ] Present the final comparison and main findings.

Their main contribution is determining whether the experimental conclusions are fair and reliable.